MAT 3007 – Optimization

Projected Gradient Descent

Lecture 25

Instructor: Junchi Yang

CUHK-Shenzhen

Recap

Newton's Method in High Dimensional Case

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top (\boldsymbol{x} - \boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k)$$

Newton's Method in High Dimensional Case

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top(\boldsymbol{x}-\boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x}-\boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x}-\boldsymbol{x}^k)$$

Newton's Method Iteration:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1}\nabla f(\boldsymbol{x}^k).$$

This corresponds to the minimizer of the quadratic approximation, provided $\nabla^2 f(\boldsymbol{x}^k)$ is positive definite.

Newton's Method in High Dimensional Case

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top(\boldsymbol{x}-\boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x}-\boldsymbol{x}^k)^\top\nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x}-\boldsymbol{x}^k)$$

Newton's Method Iteration:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1}\nabla f(\boldsymbol{x}^k).$$

This corresponds to the minimizer of the quadratic approximation, provided $\nabla^2 f(\boldsymbol{x}^k)$ is positive definite.

Search Direction (Newton's Direction):

$$\boldsymbol{d}^k = -(\nabla^2 f(\boldsymbol{x}^k))^{-1}\nabla f(\boldsymbol{x}^k).$$

Newton's Method in High Dimensional Case

We can also consider the nonlinear equation $\nabla f(\boldsymbol{x}) = \mathbf{0}$.

Newton's Method in High Dimensional Case

We can also consider the nonlinear equation $\nabla f(\boldsymbol{x}) = \mathbf{0}$.

Using a Taylor expansion at $\boldsymbol{x}^k$, we have

$$\nabla f(\boldsymbol{x}) \approx \nabla f(\boldsymbol{x}^k) + \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k) =: q_k(\boldsymbol{x}).$$

Newton's Method in High Dimensional Case

We can also consider the nonlinear equation $\nabla f(\boldsymbol{x}) = \mathbf{0}$.

Using a Taylor expansion at $\boldsymbol{x}^k$, we have

$$\nabla f(\boldsymbol{x}) \approx \nabla f(\boldsymbol{x}^k) + \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k) =: q_k(\boldsymbol{x}).$$

The solution to $q_k(\boldsymbol{x}) = \mathbf{0}$ is (when$\nabla^2 f(\boldsymbol{x})$ is invertible)

$$\boldsymbol{x} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k).$$

One-Step Convergence for Quadratic Functions

$$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x} + c,$$

where $\boldsymbol{A}$ is symmetric and positive definite, $\boldsymbol{b} \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

Properties:

- Gradient: $\nabla f(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}$,
- Hessian: $\nabla^2 f(\boldsymbol{x}) = \boldsymbol{A}$.

Newton's Update Rule:

$$\begin{aligned}\boldsymbol{x}_{k+1} &= \boldsymbol{x}_k - (\nabla^2 f(\boldsymbol{x}_k))^{-1} \nabla f(\boldsymbol{x}_k) \\ &= \boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \boldsymbol{A}^{-1}(\boldsymbol{A}\boldsymbol{x}_k - \boldsymbol{b}) = \boldsymbol{A}^{-1}\boldsymbol{b}.\end{aligned}$$

Result: Newton's method finds the exact minimizer $\boldsymbol{x}^* = \boldsymbol{A}^{-1}\boldsymbol{b}$ in **one step**, because:

- The quadratic approximation is exact for $f(\boldsymbol{x})$,

Theorem: Convergence of Newton's Method

Let f be a twice continuously differentiable function over $\mathbb{R}^n$. Assume that

- there exists $\mu > 0$ that $\nabla^2 f(\boldsymbol{x}) \succeq \mu \boldsymbol{I}$ for any $\boldsymbol{x}$,
- there exists $L > 0$ that $\|\nabla^2 f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for all $\boldsymbol{x}, \boldsymbol{y}$.

Theorem: Convergence of Newton's Method

Let f be a twice continuously differentiable function over $\mathbb{R}^n$. Assume that

- there exists $\mu > 0$ that $\nabla^2 f(\boldsymbol{x}) \succeq \mu \boldsymbol{I}$ for any $\boldsymbol{x}$,
- there exists $L > 0$ that $\|\nabla^2 f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for all $\boldsymbol{x}, \boldsymbol{y}$.

Let $\{\boldsymbol{x}^k\}_k$ be the sequence generated by Newton's method, and let $\boldsymbol{x}^*$ be the unique minizer. Then

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \frac{L}{2\mu}\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2$$

quadratic convergence

Theorem: Convergence of Newton's Method

Let f be a twice continuously differentiable function over $\mathbb{R}^n$. Assume that

- there exists $\mu > 0$ that $\nabla^2 f(\boldsymbol{x}) \succeq \mu \boldsymbol{I}$ for any $\boldsymbol{x}$,
- there exists $L > 0$ that $\|\nabla^2 f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for all $\boldsymbol{x}, \boldsymbol{y}$.

Let $\{\boldsymbol{x}^k\}_k$ be the sequence generated by Newton's method, and let $\boldsymbol{x}^*$ be the unique minizer. Then

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \frac{L}{2\mu}\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2$$

In addition, if $\|\boldsymbol{x}^0 - \boldsymbol{x}^*\| \leq \frac{\mu}{L}$, then local convergence

$$\|\boldsymbol{x}^k - \boldsymbol{x}^*\| \leq \frac{2\mu}{L}\left(\frac{1}{2}\right)^{2^k}$$

.

Convergence Newton's Method:

- "Hessian uniformly positive definite + Lipschitz continuous" and "$\boldsymbol{x}^0$ close to $\boldsymbol{x}^*$"

$$\Longrightarrow \quad \{\boldsymbol{x}^k\}_k \text{ converges quadratically to } \boldsymbol{x}^*$$

- **Proof:** See Theorem 5.2 in Beck: "Introduction to Nonlinear Optimization".

Step Length

Newton's method may not converge unless the initial point is sufficiently close to a solution.

Step Length

Newton's method may not converge unless the initial point is sufficiently close to a solution.

One way to ensure convergence is to use a step size parameter α_k

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$$

where $\boldsymbol{d}^k = -(\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k)$ is Newton's direction.

- We can use backtracking line search to determine α_k.
- $\boldsymbol{d}^k$ is a descent direction if f is strongly convex and $\nabla f(\boldsymbol{x}) \neq \boldsymbol{0}$.

Complete Procedure of Newton's Method

(Damped) Newton Method

1. Initialization: Select an initial point $\boldsymbol{x}^0 \in \mathbb{R}^n$.

For $k = 0, 1, \ldots$:

2. If $\|\nabla f(\boldsymbol{x}^k)\| \leq \varepsilon$, then STOP and $\boldsymbol{x}^k$ is the output.
3. Compute the Newton's direction $\boldsymbol{d}^k$ which is the solution of the linear system

$$\nabla^2 f(\boldsymbol{x}^k)\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k).$$

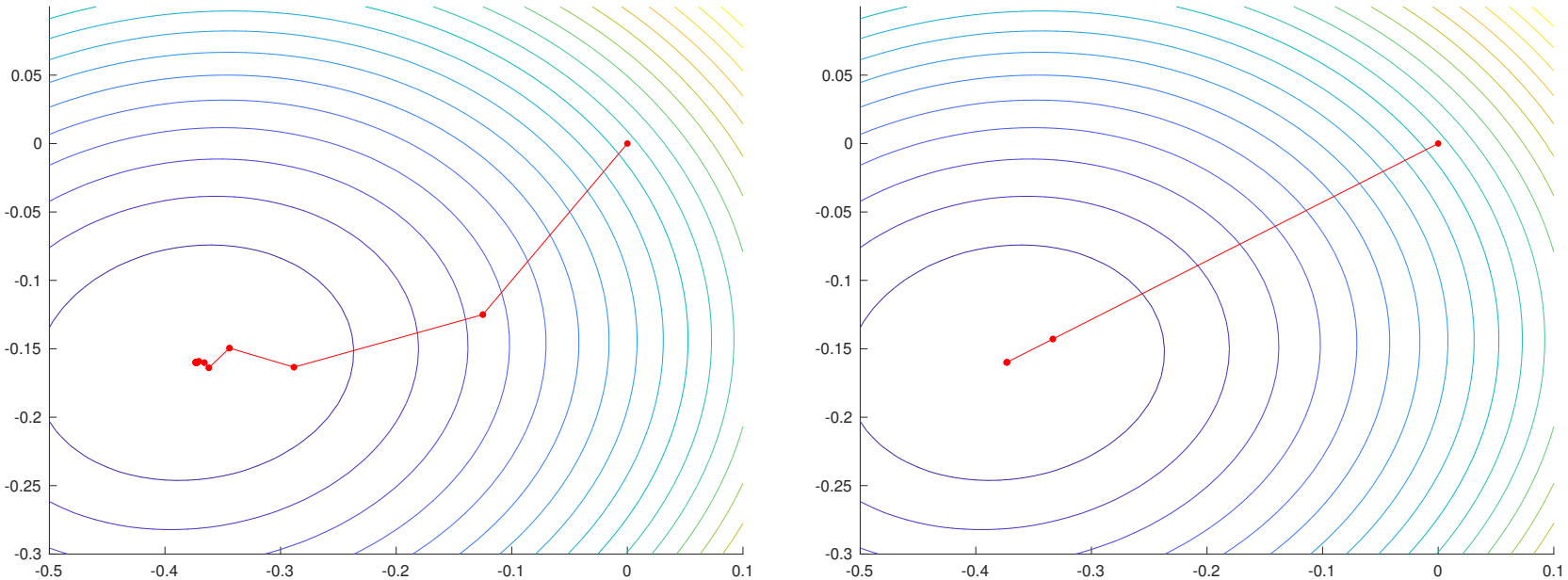
4. Newton's method: $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \boldsymbol{d}^k$
   Damped Newton's method: $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$, where $\alpha_k \leq 1$ is decided by the step size strategy, such as backtracking line search.

Example

Minimize

$$f(\boldsymbol{x}) = \exp(x_1 + x_2) + x_1^2 + 3x_2^2 - x_1 x_2$$

using the gradient method and Newton's method.



- **Left:** Gradient method (Armijo). **Right:** Newton's method.

Newton's Method is Fast, But ...

Challenges of Newton's Method:

- **Hessian Computation:** Requires $\nabla^2 f(\boldsymbol{x})$ at each iteration, computationally expensive for high-dimensional problems.
- **Memory:** Storing the Hessian matrix requires $O(n^2)$ space, significantly more than the $O(n)$ for gradient descent.
- **Matrix Operations:** Each iteration involves costly matrix inversions or solving linear systems.

Newton's Method is Fast, But ...

Challenges of Newton's Method:

- **Hessian Computation:** Requires $\nabla^2 f(\boldsymbol{x})$ at each iteration, computationally expensive for high-dimensional problems.
- **Memory:** Storing the Hessian matrix requires $O(n^2)$ space, significantly more than the $O(n)$ for gradient descent.
- **Matrix Operations:** Each iteration involves costly matrix inversions or solving linear systems.

Partial Solutions

- **Quasi-Newton Method:** Instead of calculating $\nabla^2 f(\boldsymbol{x})$ explicitly, approximate it using past gradients.
- **Inexact Newton Methods:** Solve the Newton system approximately.
- Compute Hessian only periodically.

Trade-off between Newton and Gradient Method

- Gradient method is simple and reasonably fast in each iteration and needs less storage and computation.
- Newton's method takes much less iterations to converge. However, each iteration is more expensive in both computation and storage space.

There are many numerical improvements that one can make but the above main tradeoff always remains:

- Whether the gradient method or Newton's method should be used depends largely on the problem at hand.

Algorithms for Constrained Problems

Agenda

In the following, we will extend our discussions to constrained optimization problems.

- We will introduce one algorithm for such problems — the projected gradient method.

Constrained Optimization

We consider the following constrained optimization problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \Omega. \end{aligned}$$

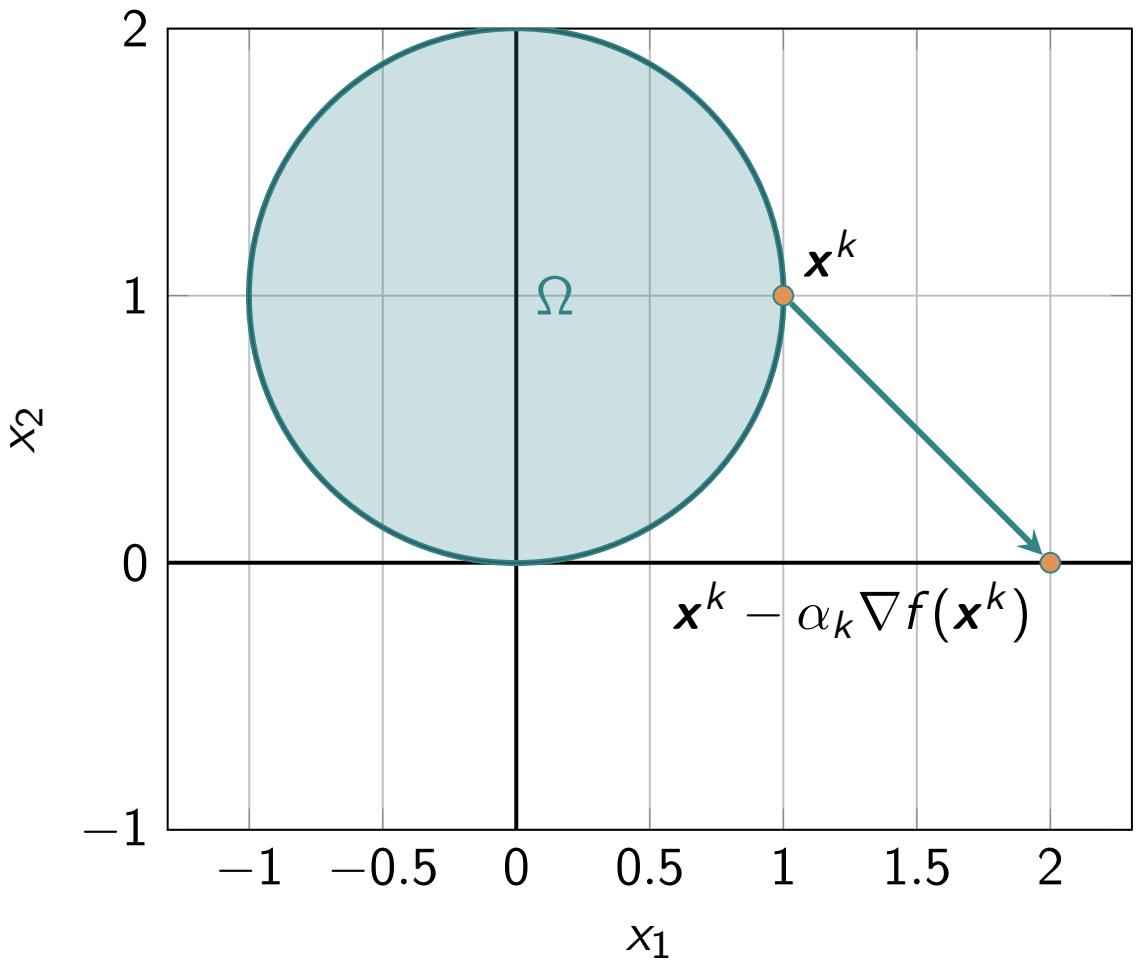
In the methods for unconstrained problems, the main idea is:

- At each $\boldsymbol{x}^k$, compute a descent direction $\boldsymbol{d}^k$.
- Then find an appropriate step size α_k and update $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.

However,Under Additional Constraints:

- $\boldsymbol{x}^{k+1}$ can be infeasible.

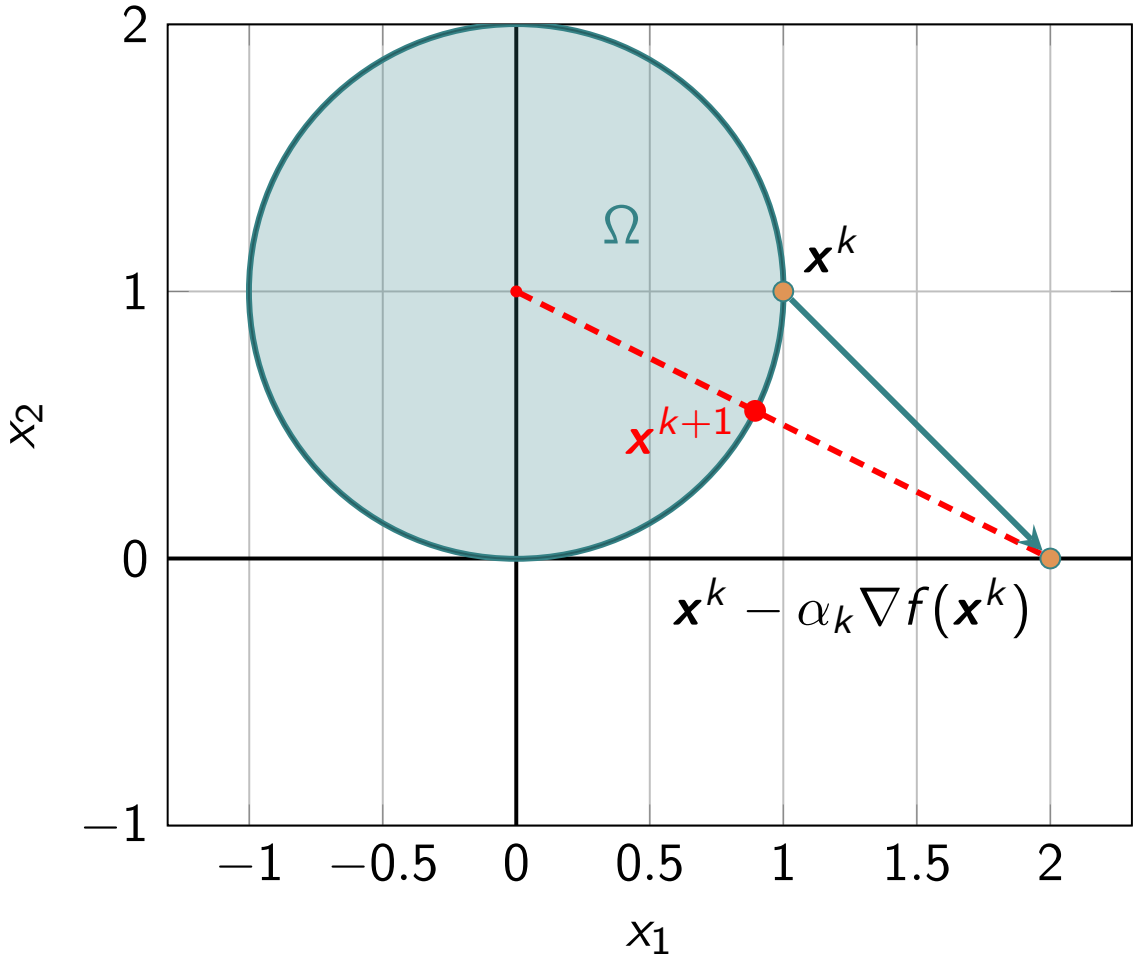
Illustration



- One solution to this problem is to use the projected gradient method.

Main Idea of the Projected Gradient Method

Project the gradient step back onto the feasible set:



- We can use the general framework by maintaining feasibility.

Projections

Definition: Euclidean Projections

Let $\Omega \subset \mathbb{R}^n$ be a nonempty, closed, convex set.

The (Euclidean) projection of $\boldsymbol{x}$ onto Ω is defined as the unique minimizer $\boldsymbol{y}^*$ of the constrained optimization problem:

$$\min_{\boldsymbol{y}} \ \frac{1}{2}\|\boldsymbol{x}-\boldsymbol{y}\|^2 \quad \text{s.t.} \quad \boldsymbol{y} \in \Omega$$

and we write $\boldsymbol{y}^* = \mathcal{P}_\Omega(\boldsymbol{x})$.

This is a result of strong convexity of the objective function (we didn't cover this in class).

Observation:

Definition: Euclidean Projections

Let $\Omega \subset \mathbb{R}^n$ be a nonempty, closed, convex set.

The (Euclidean) projection of $\boldsymbol{x}$ onto Ω is defined as the unique minimizer $\boldsymbol{y}^*$ of the constrained optimization problem:

$$\min_{\boldsymbol{y}} \ \frac{1}{2}\|\boldsymbol{x}-\boldsymbol{y}\|^2 \quad \text{s.t.} \quad \boldsymbol{y} \in \Omega$$

and we write $\boldsymbol{y}^* = \mathcal{P}_\Omega(\boldsymbol{x})$.

Observation:

- The projection $\boldsymbol{y}^* = \mathcal{P}_\Omega(\boldsymbol{x})$ is the point in Ω that has the minimum distance to $\boldsymbol{x}$.

Example — I: Linear Constraints

We first consider the simple case where Ω consists of linear equality constraints:

$$\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\},$$

where $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$ are given.

Example — I: Linear Constraints

We first consider the simple case where Ω consists of linear equality constraints:

$$\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\},$$

where $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$ are given.

Euclidean Projection:

- Suppose that $\boldsymbol{A}$ has full row rank, then it holds that:

$$\mathcal{P}_\Omega(\boldsymbol{x}) = \boldsymbol{x} - \boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{A}^\top)^{-1}[\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}].$$

Proof: By definition of projection,

$$\min_y \frac{1}{2}\|y-x\|^2 \quad \text{s.t} \quad \underline{y\in\Omega} \rightarrow Ay=b.$$

Lagrangian: $L(y,\mu) = \frac{1}{2}\|y-x\|^2 + \mu^T(Ay-b)$

KKT: $\nabla_y L(y,\mu) = y-x+A^T\mu = 0$ ①

$Ay = b.$ ②

Assume A is <u>full-row rank</u>, then AA^T is invertible.

From ①, we have $y = -A^T\mu + x \Rightarrow Ay = Ax - AA^T\mu.$

Combining with ② $b = Ay = Ax - AA^T\mu \Rightarrow \mu = [AA^T]^{-1}(Ax-b)$

$$P_\Omega(x) = y = -A^T\mu + x = x - A^T[AA^T]^{-1}(Ax-b)$$ □

Example — II: Box Constraints

Suppose that Ω is given by box constraints:

$$\Omega = \{\boldsymbol{x} \in \mathbb{R}^n : x_i \in [a_i, b_i],\ \forall\ i\} = [\boldsymbol{a}, \boldsymbol{b}],$$

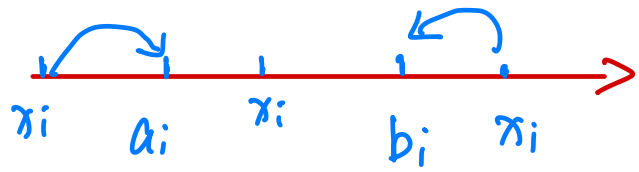
where $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$, $\boldsymbol{a} \leq \boldsymbol{b}$, are given.

Example — II: Box Constraints

Suppose that Ω is given by box constraints:

$$\Omega = \{\boldsymbol{x} \in \mathbb{R}^n : x_i \in [a_i, b_i],\ \forall\ i\} = [\boldsymbol{a}, \boldsymbol{b}],$$

where $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$, $\boldsymbol{a} \leq \boldsymbol{b}$, are given.



Euclidean Projection:

- The projection onto Ω can be computed as follows:

$$[\mathcal{P}_\Omega(\boldsymbol{x})]_i = \mathcal{P}_{[a_i,b_i]}(x_i) = \max\{\min\{x_i, b_i\}, a_i\} \quad \forall\ i.$$

(Exercise: verify this is proj. onto the box constraint)

$$= \begin{cases} x_i & \text{if } x_i \in [a_i, b_i] \\ b_i & \text{if } x_i > b_i \\ a_i & \text{if } x_i < a_i \end{cases}$$

Example — III: Ball Constraints

Suppose that Ω is a Euclidean ball with radius $r > 0$ and center $\boldsymbol{m} \in \mathbb{R}^n$, i.e.:

$$\Omega = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{m}\| \leq r\}.$$

Example — III: Ball Constraints

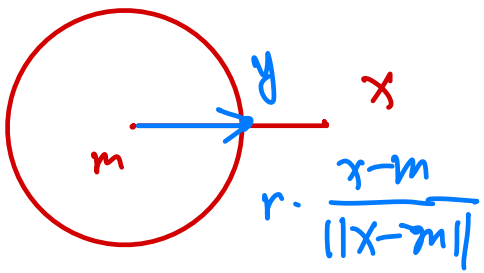
Suppose that Ω is a Euclidean ball with radius $r > 0$ and center $\boldsymbol{m} \in \mathbb{R}^n$, i.e.:

$$\Omega = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{m}\| \leq r\}.$$

Euclidean Projection:

- The projection onto Ω can be computed as follows:

$$\mathcal{P}_\Omega(\boldsymbol{x}) = \begin{cases} \boldsymbol{x} & \text{if } \|\boldsymbol{x} - \boldsymbol{m}\| \leq r, \\ \boldsymbol{m} + \frac{r}{\|x-m\|}(\boldsymbol{x} - \boldsymbol{m}) & \text{if } \|\boldsymbol{x} - \boldsymbol{m}\| \geq r. \end{cases}$$



proof: By def. of projection, consider

$$\min_y \ \frac{1}{2}\|y-x\|^2 \quad \text{s.t} \quad \|y-m\| \le r$$

rewrite as $\min_y \ \frac{1}{2}\|y-x\|^2 \quad \text{s.t} \ \frac{1}{2}\|y-m\|^2 \le \frac{1}{2}r^2$

Denote $z = y-m$, then

$$\min_z \ \frac{1}{2}\|z+m-x\|^2 \quad \text{s.t} \ \frac{1}{2}\|z\|^2 \le \frac{1}{2}r^2$$

Lagrangian: $L(z,\lambda) = \frac{1}{2}\|z+m-x\|^2 + \lambda\left[\frac{1}{2}\|z\|^2 - \frac{1}{2}r^2\right]$

KKT condition: $z+m-x+\lambda z = 0$

$$\|z\| \le r, \quad \lambda \ge 0, \quad \lambda\left(\|z\|^2 - r^2\right) = 0$$

Case 1: $\|z\|^2 - r^2 < 0 \xrightarrow{\text{complementarity}} \lambda = 0$ and by the main condition

$z = \frac{1}{1+\lambda}(x-m) = x-m$ (for this case $\|x-m\| < r$)

Case 2 : $\|z\|^2 - r^2 = 0$

$$r = \|z\| = \left\| \frac{1}{1+\lambda}(x-m) \right\| = \frac{1}{1+\lambda} \|x-m\|$$

$$\Rightarrow \frac{1}{1+\lambda} = \frac{r}{\|x-m\|} \quad \text{and}$$

$$z = \frac{1}{1+\lambda}(x-m) = r \frac{x-m}{\|x-m\|}$$

(for this case, $\|x-m\| = (1+\lambda) r \geq r$).

To conclude,

$$z^* = \begin{cases} x-m, & \text{if } \|x-m\| < r \\ r \cdot \frac{x-m}{\|x-m\|} & \text{if } \|x-m\| \geq r \end{cases}$$

$$y^* = z^* + m$$

Remark: Projections

Observation:

- For many sets, explicit formulae for the projections can be derived.
- Main Tool: KKT-conditions and convexity.
- Many more interesting projections can be expressed efficiently:

$$\mathcal{P}_{\Delta_n}(\boldsymbol{x}), \quad \mathcal{P}_{\mathbb{S}^n_+}(\boldsymbol{X}), \quad \dots$$

  where $\Delta_n := \{\boldsymbol{x} : \mathbb{1}^\top \boldsymbol{x} = 1, \boldsymbol{x} \geq \boldsymbol{0}\}$ is the n-simplex and $\mathbb{S}^n_+$ is the set of positive semidefinite matrices.
- In general: an optimization problem needs to be solved to obtain $\mathcal{P}_\Omega(\boldsymbol{x})$.

Properties of Projections and Descent Directions

Optimization Problems with Convex Constraints

Constrained minimization problems of the form:

$$\min_{\boldsymbol{y}} \; f(\boldsymbol{y}) \quad \text{s.t.} \quad \boldsymbol{y} \in \Omega, \tag{1}$$

where $\Omega \subset \mathbb{R}^n$ is a convex and closed set.

Optimization Problems with Convex Constraints

Constrained minimization problems of the form:

$$\min_{\boldsymbol{y}} \; f(\boldsymbol{y}) \quad \text{s.t.} \quad \boldsymbol{y} \in \Omega, \tag{1}$$

where $\Omega \subset \mathbb{R}^n$ is a convex and closed set.

Theorem: FONC for Problems with Convex Constraints

Let f be continuously differentiable on an open set that contains the convex, closed set $\Omega \subset \mathbb{R}^n$. Let $\boldsymbol{y}^* \in \Omega$ be a local minimizer of (1), then:

$$\nabla f(\boldsymbol{y}^*)^\top (\boldsymbol{y} - \boldsymbol{y}^*) \geq 0, \quad \forall\, \boldsymbol{y} \in \Omega. \tag{2}$$

Optimization Problems with Convex Constraints

Constrained minimization problems of the form:

$$\min_{\boldsymbol{y}} \; f(\boldsymbol{y}) \quad \text{s.t.} \quad \boldsymbol{y} \in \Omega, \tag{1}$$

where $\Omega \subset \mathbb{R}^n$ is a convex and closed set.

Theorem: FONC for Problems with Convex Constraints

Let f be continuously differentiable on an open set that contains the convex, closed set $\Omega \subset \mathbb{R}^n$. Let $\boldsymbol{y}^* \in \Omega$ be a local minimizer of (1), then:

$$\nabla f(\boldsymbol{y}^*)^\top (\boldsymbol{y} - \boldsymbol{y}^*) \geq 0, \quad \forall\, \boldsymbol{y} \in \Omega. \tag{2}$$

- If f is convex, then $\boldsymbol{y}^* \in \Omega$ is global minimizer of (1) iff the FONC is satisfied.
- A point satisfying (2) is called a stationary point of (1).

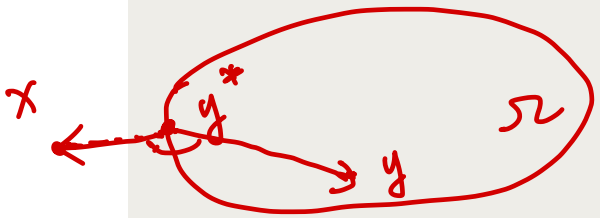
Properties of the Projection Mapping

Projection Theorem

Let Ω be a nonempty, closed, and convex set. Then:

- A point $\boldsymbol{y}^*$ is the projection of $\boldsymbol{x}$ onto Ω, i.e., $\boldsymbol{y}^* = \mathcal{P}_\Omega(\boldsymbol{x})$, if and only if

$$(\boldsymbol{y}^* - \boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{y}^*) \geq 0, \quad \forall\, \boldsymbol{y} \in \Omega.$$



$\min_y \frac{1}{2}\|y-x\|^2$ s.t. $y \in \Omega$

By FONC, $(y^*-x)^\top (y-y^*) \geq 0$

Properties of the Projection Mapping

Projection Theorem

Let Ω be a nonempty, closed, and convex set. Then:

- A point $\boldsymbol{y}^*$ is the projection of $\boldsymbol{x}$ onto Ω, i.e., $\boldsymbol{y}^* = \mathcal{P}_\Omega(\boldsymbol{x})$, if and only if

$$(\boldsymbol{y}^* - \boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{y}^*) \geq 0, \quad \forall\, \boldsymbol{y} \in \Omega.$$

- The mapping $\mathcal{P}_\Omega : \mathbb{R}^n \to \mathbb{R}^n$ is Lipschitz continuous with constant $L = 1$. $\Leftrightarrow \| P_\Omega(x) - P_\Omega(y) \| \leq \| x - y \|$

Properties of the Projection Mapping

Projection Theorem

Let Ω be a nonempty, closed, and convex set. Then:

- A point $\boldsymbol{y}^*$ is the projection of $\boldsymbol{x}$ onto Ω, i.e., $\boldsymbol{y}^* = \mathcal{P}_\Omega(\boldsymbol{x})$, if and only if

$$(\boldsymbol{y}^* - \boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{y}^*) \geq 0, \quad \forall\ \boldsymbol{y} \in \Omega.$$

- The mapping $\mathcal{P}_\Omega : \mathbb{R}^n \to \mathbb{R}^n$ is Lipschitz continuous with constant $L = 1$.
- The vector $\boldsymbol{x}^*$ is a stationary point of (1) if and only if

$$\boldsymbol{x}^* - \mathcal{P}_\Omega(\boldsymbol{x}^* - \lambda \nabla f(\boldsymbol{x}^*)) = \boldsymbol{0} \quad \text{for any } \lambda > 0.$$

x^* is fixed point for $P_\Omega(\cdot - \lambda \nabla f(\cdot))$

Motivation & Strategy:

- Setting $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)$ might likely generate infeasible iterates: $\boldsymbol{x}^{k+1} \notin \Omega$.
- Idea: We project the step $\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)$ back onto Ω:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

Remark: if $x^{k+1} = x^k \Rightarrow x^k$ is a stationary point.

- The algorithm will converge with proper assumptions and step sizes.

Line Search:

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Line Search:

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Observation:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) = \boldsymbol{x}^k + [\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k].$$

Line Search:

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Observation:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) = \boldsymbol{x}^k + [\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k].$$

- Consider $\boldsymbol{d}^k = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k$ as a search direction. Introduce another step size α_k:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k = (1 - \alpha_k)\boldsymbol{x}^k + \alpha_k \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

Line Search:

If λ_k is determined by line search, every adjustment of λ_k requires to reevaluate $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k))$. This can be expensive.

Observation:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) = \boldsymbol{x}^k + [\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k].$$

- Consider $\boldsymbol{d}^k = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k$ as a search direction. Introduce another step size α_k:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k = (1 - \alpha_k)\boldsymbol{x}^k + \alpha_k \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

$\boldsymbol{x}^k \in \Omega$ (under the first term); $\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) \in \Omega$ (under the second term)

$\rightsquigarrow$ If $\alpha_k \in [0, 1]$, then the convexity of Ω implies that $\boldsymbol{x}^{k+1}$ will be feasible if $\boldsymbol{x}^k \in \Omega$.

Descent Directions

Let $\boldsymbol{x} \in \Omega$ and $\lambda > 0$ be given. If $\boldsymbol{x}$ is not a stationary point of (1), then the direction $\boldsymbol{d} := \mathcal{P}_\Omega(\boldsymbol{x} - \lambda \nabla f(\boldsymbol{x})) - \boldsymbol{x}$ is a descent direction and it holds that

$$\nabla f(\boldsymbol{x})^\top \boldsymbol{d} \leq -\frac{1}{\lambda}\|\boldsymbol{d}\|^2 < 0.$$

- We can now reuse similar techniques like backtracking to generate step sizes.
- We can use $\|\mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k\| \leq \varepsilon$ as stopping criterion.

$\rightsquigarrow$ The only difference between the gradient and the projected method is the search direction and stopping condition.

Projected Gradient Method with Backtracking Line Search

1. Initialization: Choose an initial point $\boldsymbol{x}^0 \in \Omega$ and $\sigma, \gamma \in (0,1)$.

For $k = 0, 1, \ldots$:

2. Select $\lambda_k > 0$ and compute $\nabla f(\boldsymbol{x}^k)$ and the new direction $\boldsymbol{d}^k = \mathcal{P}_\Omega(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)) - \boldsymbol{x}^k$.
3. If $\|\boldsymbol{d}^k\| \leq \lambda_k \varepsilon$, then STOP and $\boldsymbol{x}^k$ is the output.
4. Choose a maximal step size $\alpha_k \in \{1, \sigma, \sigma^2, \ldots\} \subset (0,1]$ that satisfies the Armijo condition

$$f(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k) - f(\boldsymbol{x}^k) \leq \gamma \alpha_k \cdot \nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k.$$

5. Set $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.

Theorem: Convergence Projected Gradient Method



With backtracking line search

Assumptions: f is cont. diff., Ω is nonempty, convex, and closed and the step sizes $\{\lambda_k\}_k$ are bounded:

$$0 < \underline{\lambda} \leq \lambda_k \leq \overline{\lambda} \quad \forall\ k.$$

$\rightsquigarrow$ Every accumulation point of $\{\boldsymbol{x}^k\}$ is a stationary point.

Theorem: Convergence Projected Gradient Method



Assumptions: f is cont. diff., Ω is nonempty, convex, and closed and the step sizes $\{\lambda_k\}_k$ are bounded:

$$0 < \underline{\lambda} \leq \lambda_k \leq \overline{\lambda} \quad \forall\, k.$$

for constraint problem

$\rightsquigarrow$ Every accumulation point of $\{\boldsymbol{x}^k\}$ is a stationary point.

$\|\nabla f(x) - \nabla f(y)\| \leq L\|x-y\| \quad \forall x, y$

If ∇f is additionally Lipschitz continuous, we obtain:

- If f is convex, we converge to global solutions of the problem.
- If $\lambda_k \leq \frac{2(1-\gamma)}{L}$, then $\alpha_k = 1$ will be accepted as step size.
- If f is strongly convex, then $\{\boldsymbol{x}^k\}_k$ converges linearly to a global solution $\boldsymbol{x}^* \in \Omega$.

Remark: the convergence rate is also affected by the condition number, like gradient descent.

Example — I

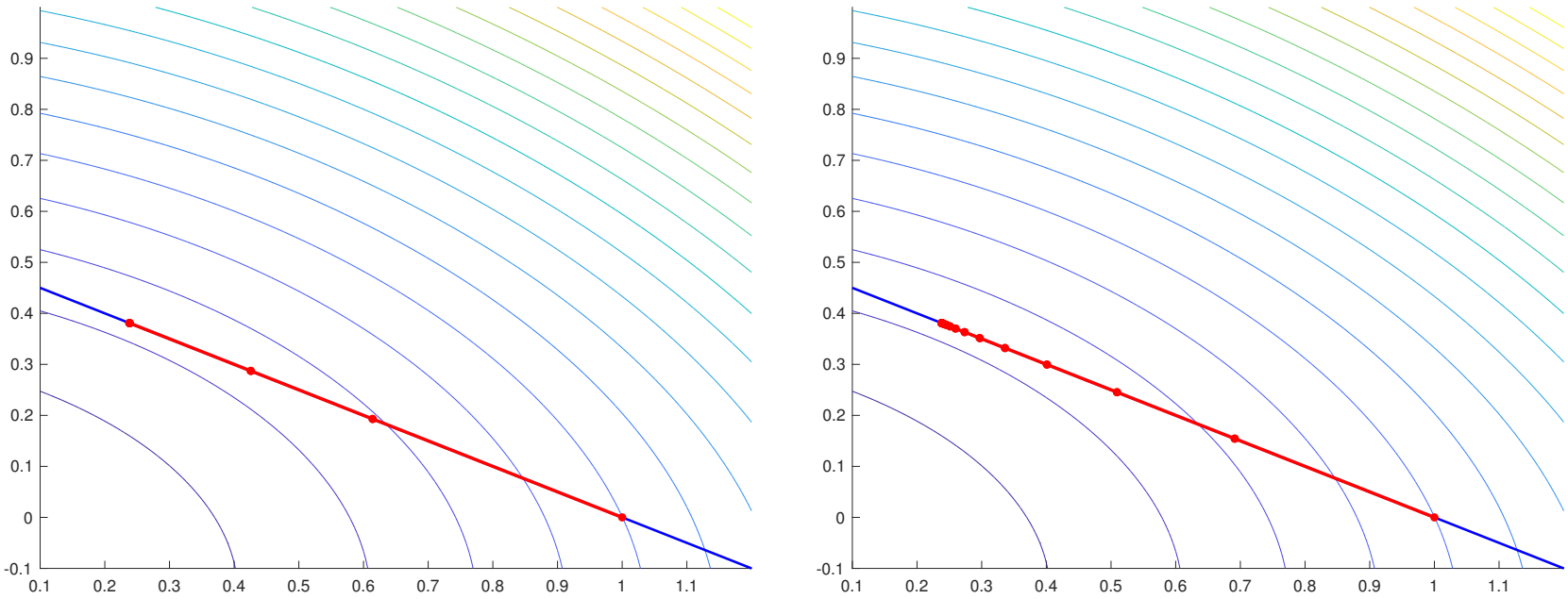
$$\begin{aligned} \text{minimize} \quad & e^{x_1+x_2} + x_1^2 + 3x_2^2 - x_1 x_2 \\ \text{subject to} \quad & x_1 + 2x_2 = 1 \end{aligned}$$

Setting $\boldsymbol{A} = [1, 2]$, $b = 1$, and $\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = b\}$, we have

$$\mathcal{P}_\Omega(\boldsymbol{x}) = \begin{bmatrix} 4/5 & -2/5 \\ -2/5 & 1/5 \end{bmatrix} x + \begin{bmatrix} 1/5 \\ 2/5 \end{bmatrix}$$

We use the initial feasible point $\boldsymbol{x}^0 = (1, 0)^\top$ and $\lambda_k = 1$ and $\lambda_k = 0.1$.

Example — I: Results



- **Left:** $\lambda_k = 1$. **Right:** $\lambda_k = 0.1$.

Today's lecture

- Projected Gradient Descent

Next lecture

- Integer Programming

Questions?